

LIYING LI, Ph.D.

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EDUCATION

Ph.D., Environmental Science — University of California, Merced	2024
M.S., Sustainability Project Management — Northwestern University	2015
M.Sc., Environmental Policy and Regulation — London School of Economics and Political Science	2012
B.A., Environmental Engineering — Dalian University of Technology	2011

ACADEMIC APPOINTMENTS

Postdoctoral Researcher — National Center for Ecological Analysis and Synthesis (NCEAS), UC Santa Barbara	2024 – 2026
Graduate Research Assistant — University of California, Merced	2021 – 2024
Associate Researcher — School of Environment, Tsinghua University	2016 – 2019
Research Assistant — School of Environment, Tsinghua University	2012 – 2013

METHODOLOGICAL EXPERTISE

Deep Learning Architectures Hierarchical and generative models with domain-informed priors; graph neural networks; LSTM and sequence models; mixture-of-experts for multi-scale processes; hurdle and zero-inflated architectures for sparse count data; spatiotemporal adaptive ensembles.

Bayesian & Statistical Inference Hierarchical and multilevel models; prior specification from domain structure; posterior uncertainty quantification and calibration; species distribution and population models; step-selection and movement models; geostatistics and spatial statistics.

Interpretability & Scientific ML Variable-importance and driver-attribution analysis; encoding physical and ecological structure as inductive bias; out-of-sample validation on large, unevenly sampled observational datasets; structural decomposition and attribution.

Data management & Computation R and Python; integration of heterogeneous sources (satellite remote sensing, citizen-science archives, field monitoring, telemetry, administrative panels); reproducible analysis pipelines; Git/GitHub; public release of code and model implementations.

Application Domains Biodiversity under climate change and extreme events; coastal, estuarine, and fisheries systems; coupled food–energy–water systems; multi-objective conservation planning; environmental economics and policy analysis.

PEER-REVIEWED PUBLICATIONS

2026

1. Li, L., Bai, J., Sun, S., Zuzuarregui, M., & Wang, Z. The risk of birds' functional diversity loss with climate change uncovered using deep learning hurdle models. *Ecological Indicators*, 190, 115321. [[hurdle deep learning](#)]
2. Li, L., Zuzuarregui, M., Bai, J., Sun, S., Chen, Y., & Wang, Z. Hurricanes drive bird displacement revealed by deep learning species distribution models. *Ecological Informatics*, 95, 103785. [[event-response modeling](#)]
3. Li, L., Hestir, E., Fink, D., Rodriguez-Flores, J. M., Viers, J. H., Medellin-Azuara, J., & Conklin, M. Nature-based solutions to reduce carbon emissions, control groundwater overdraft, and conserve avian biodiversity in California. *Environmental and Sustainability Indicators*, 101180. [[coupled systems](#)]
4. Li, L., Rodemann, J., Hayes, C., Correia, K. M., Darnell, Z., Smee, L., Alford, S., Belgrad, B., Hall, P., McDonald, A., Smith, S., & Furman, B. Variation in seagrass habitat use by fishery-important nekton across the Gulf of America revealed by hierarchical deep learning with trophic-level priors. *BioRxiv preprint*. [[hierarchical priors](#)]

2025

5. Li, L., Cole, S., Rodriguez-Flores, J. M., Hestir, E., Fink, D., Viers, J. H., Medellin-Azuara, J., Conklin, M., & Harmon, T. Synergies between agricultural production and shorebird conservation with climate change in the Central Valley, California, with optimized water allocation and multi-benefit land use. *Global Change Biology*, 31(6). [[spatial and temporal species distribution modeling](#)]
6. Li, L., Dogan, M. S., Maskey, M., et al. Optimized water allocation with managed groundwater recharge and prioritized wetland deliveries to moderate human–nature water use tradeoffs under climate change. *Journal of Hydrology: Regional Studies*, 60, 102496. [[economic optimization](#)]

2022 and earlier

7. Maskey, M. L., Dogan, M. S., Fernandez-Bou, A. S., Li, L., Guzman, A., Arnold, W., Goharian, E., Lund, J. R., & Medellin-Azuara, J. (2022). Managing aquifer recharge to overcome overdraft in the Lower American River, California, USA. *Water*, 14(6), 966.

8. Li, L. (2021). Integrating climate change impact in new building design process: A review of building life cycle carbon emission assessment methodologies. *Cleaner Engineering and Technology*, 5, 100286.
9. Li, L. (2021). Assessing climate change impacts and adaptation options of rain-fed agriculture in Africa with an integrated modeling framework. *Geo-Extreme* 2021, 203–212.
10. Li, L. (2020). The governance of low-carbon transitions in a multilevel perspective framework: How does the concept of "system transformation" work? *Energy Research Journal*, 11, 45–53.
11. Li, X., Chen, S., Li, L., Quan, X., & Zhao, H. (2014). Electrochemically enhanced adsorption of nonylphenol on carbon nanotubes: Kinetics and isotherms study. *Journal of Colloid and Interface Science*, 415, 159–164. [34 citations]
12. Wang, Y., Zhao, H., Li, L., Liu, Z., & Liang, S. (2013). Carbon dioxide emission drivers for a typical metropolis using input–output structural decomposition analysis. *Energy Policy*, 58, 312–318. [220 citations]

MANUSCRIPTS IN REVIEW AND IN PREPARATION

- Li, L., Griffin, L., Lamont, M., Kelley, K., Rodemann, J., Baustian, M., Dance, M., Ellis, R., Fogg, A., Franks, J., Gibson, D., Hendon, R., Higgs, J., Kendall, B., Powers, S., Price, M., Streich, M., & Young, J. Distributions and movement patterns of cobia (*Rachycentron canadum*) in the Gulf of America modeled using a multi-scale mixture-of-experts step-selection function. — under organizational review
- Li, L., Griffin, L., Lamont, M., Kelley, K., Rodemann, J., Baustian, M., Dance, M., Ellis, R., Fogg, A., Franks, J., Gibson, D., Hendon, R., Higgs, J., Kendall, B., Powers, S., Price, M., Streich, M., & Young, J. Modeling fish varying swimming speeds across movement states, seasons, and life stages— In preparation
- Li, L., Flaherty Walia, K., & Schram, M. Bayesian GNN-nLSTM discovers the relative importance of abiotic and biotic interactions to fish stocks. — in preparation
- Nelson, E., & Li, L. Mapping coastal species genetic diversity in California and informing ecosystem-based conservation to enhance climate change resiliency. — in preparation
- Li, L., Wang, Z., Nesslage, J., Boya Zhang. Debiasing biodiversity pattern with deep learning. — in preparation

CONFERENCE PRESENTATIONS

- Li, L. Predicting storm surge impacts on avian biodiversity using spatiotemporal adaptive ensemble deep learning models. American Geophysical Union Fall Meeting, 2025. Poster.
- Li, L. Multi-benefit conservation planning for balancing competing land and water use for human and biodiversity conservation. American Geophysical Union Fall Meeting, 2024.
- Li, L., & Medellin-Azuara, J. Integrated modeling framework for shorebird population changes under land-use and climate change using remotely sensed and citizen-science data for Central Valley farmlands, California. American Geophysical Union Fall Meeting, 2023. Oral.
- Li, L., Cole, S., Hestir, E. L., & Medellin-Azuara, J. Identifying priority areas for non-breeding shorebird habitat provisioning from agricultural land: Conserving water, biodiversity, and agriculture. International Geoscience and Remote Sensing Symposium (IGARSS), 2023. Oral.
- Li, L., & Medellin-Azuara, J. Co-benefits of managed aquifer recharge in California: Integrated assessment of climate and land-use change impacts on agriculture with spatially explicit ecosystem service analysis. American Geophysical Union Fall Meeting, 2021. Poster.
- Qing, C., Miao, C., Li, L., & Guo, P. Analysis of the investment need for municipal wastewater treatment facility construction in China during the 13th Five-Year Plan period. ICSI Conference, 2016. Poster.

SOFTWARE AND OPEN RESEARCH

Model implementations and analysis code from published work are released publicly at github.com/lily6966, with project documentation and manuscript preprints at lily6966.github.io. Architectures released to date include hurdle and generative deep learning models for sparse ecological counts, hierarchical models with domain-informed priors, and Bayesian graph/sequence models for driver attribution.

FELLOWSHIPS AND FUNDED RESEARCH

Gulf Ecosystem Initiative Postdoctoral Fellowship	2024
Labor & Automation in California Agriculture Summer Research Fellow (Grant 21-0557-000-SO)	2024
UC Environmental System Bobcat Summer Fellowship	2021, 2023
National Institute of Food and Agriculture (2021-69012-35916) & UC Regents Summer Fellowship	2023
UC Environmental System Professional Development Fellowship	2022
Bakersfield College Faculty Diversification Fellowship	2022
NSF–USDA INFEWS Graduate Research Assistantship (Grant 1639268)	2021

TEACHING AND MENTORING

University of California, Merced*Graduate Teaching Assistant***2021 – 2024***Merced, CA*

- Engineering Economics (ENGR 065), two semesters: led weekly discussion and problem-solving sections on cost–benefit analysis, decision-making under uncertainty, discounted cash flow, and project evaluation; guided quantitative exercises and engineering case studies.
- Mentored undergraduates in statistics, programming, and environmental data analysis, supporting reproducible R and Python workflows, data visualization, and interpretation of analytical results; supervised undergraduate research projects one-on-one.

Bakersfield College*Teaching Fellow***2022 – 2023***Bakersfield, CA*

- Completed professional training in evidence-based college teaching: active learning, learning assessment, course and outcome design, inclusive practice for diverse and first-generation students; delivered practice lectures in environmental engineering and science.

SERVICE AND PROFESSIONAL ACTIVITIES

Peer reviewer: Journal of Hydrology · Global Change Biology · Biological Conservation

Member: American Geophysical Union (AGU) · American Association for the Advancement of Science (AAAS) · Ecological Society of America (ESA) · Geological Society of America (GSA) · Society for Conservation Biology (SCB)

REFERENCES

Benjamin Halpern — Director, NCEAS, University of California, Santa Barbara · halpern@nceas.ucsb.edu

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